

POLYGLOT WORKSHOP



ENVIRONMENTAL PRACTICES TO LLM DEVELOPMENT

Tutors

Nicholas Kluge and Sophia Falk



AGENDA TODAY

- ✓ Introduction to Sustainability of AI
- ✓ Methods for estimating energy consumption and carbon footprint of LLMs
- ✓ How data center location affects environmental impact
- ✓ How to implement a carbon-tracking system in your AI development workflow
- ✓ Beyond carbon: other dimensions of sustainability in AI development



INTRODUCTION TO SUSTAINABILITY OF AI



Fairwater 2 - Atlanta: Microsofts AI superfactory

November 12, 2025 // image: microsoft.com

- CLUSTERS OF HUNDREDS OF THOUSANDS OF NVIDIA GPUS,**
- FIBRE OPTIC CABLES WOULD STRETCH 4.5 TIMES AROUND THE EARTH.**
- 10 TIMES THE PERFORMANCE OF THE WORLD'S FASTEST SUPERCOMPUTER TODAY**
- APPROX. 2GW POWER CONSUMPTION**



AI SUSTAINABILITY DEBATE

01 ENERGY CONSUMPTION

02 WATER CONSUMPTION



AI SUSTAINABILITY DEBATE

01 ENERGY CONSUMPTION

data center location | energy
supply

02 WATER CONSUMPTION

chip manufacturing | data center
cooling



AI SUSTAINABILITY DEBATE

01 ENERGY CONSUMPTION

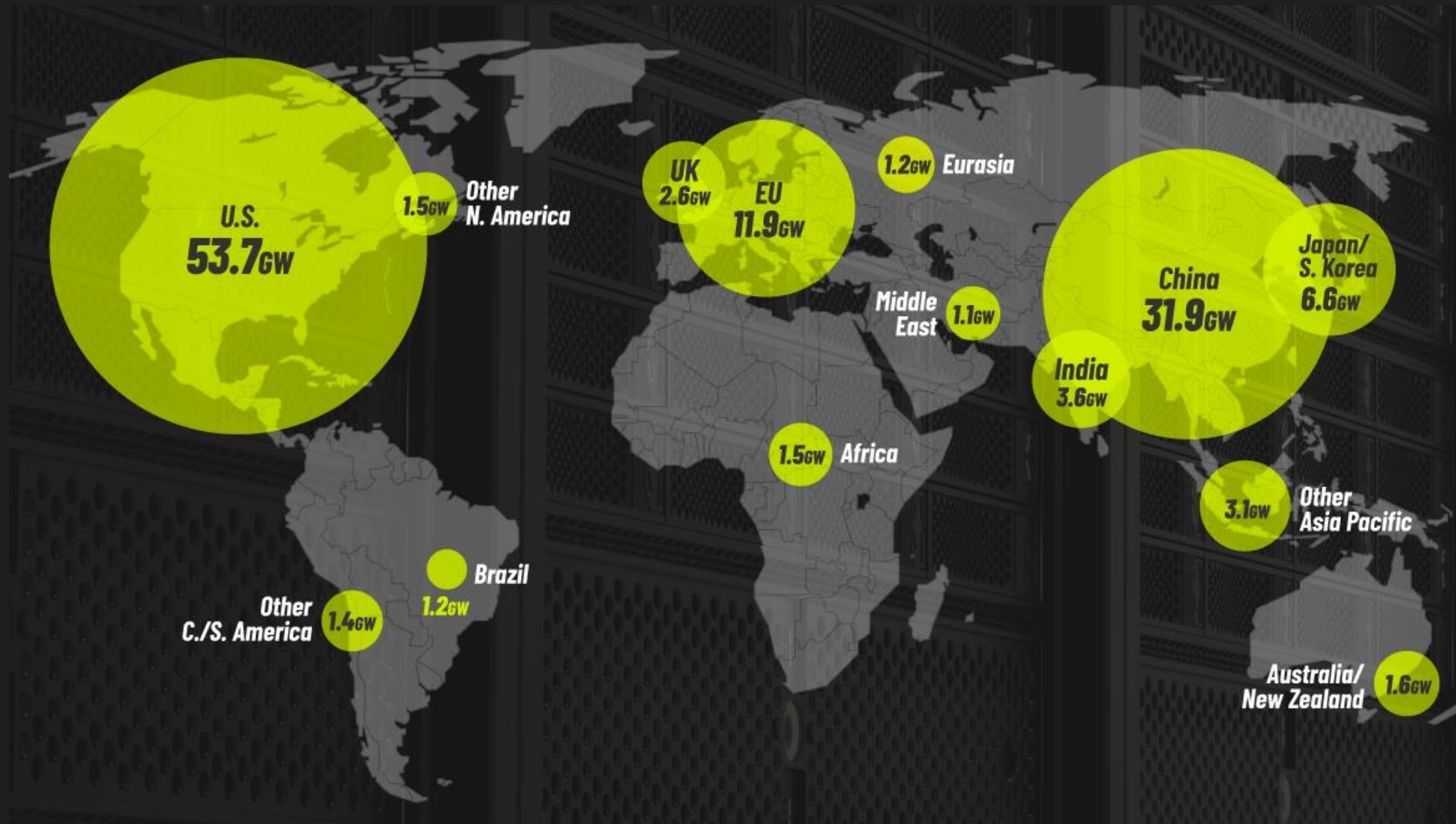
data center location | energy
supply

02 WATER CONSUMPTION

chip manufacturing | data center
cooling



DATA CAPACITY AROUND THE WORLD (July 2025)



<https://www.visualcapitalist.com/data-center-capacity-around-the-world/>

Global installed data center capacity stood at 122.2 gigawatts (GW) in 2024.

Together, the U.S. and China account for 70% of the global total.

The European Union follows next, with 11.9 GW in capacity.

AI AND WORLD ENERGY OUTLOOK



*„Data centers are **one of several** drivers of accelerated electricity demand growth in the Age of Electricity“*

Data centers accounted for **around 1.5% of the world's electricity consumption in 2024**, or 415 terawatt-hours (TWh).

By 2030:

- more than double (945 TWh) (3% global electricity demand)
- Projected 200 GW capacity
- In the US DCs use more electricity than aluminum, steel, cement, chemicals and all other energy-intensive goods combined.
- Electricity grids are already under pressure.
- Questionable how bottlenecks will be resolved (IEA, 2025).



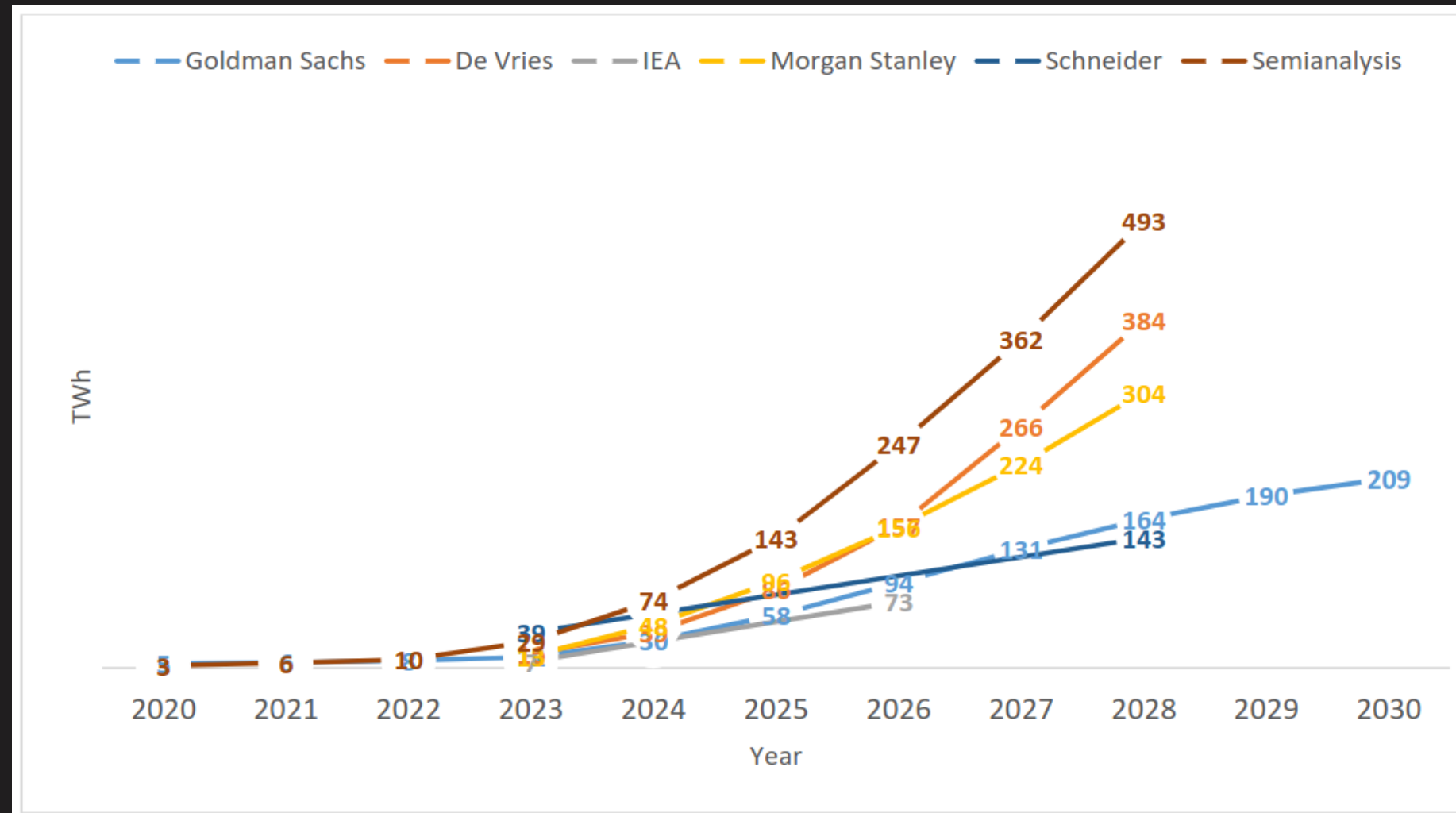
GIGAWATT AND TERAWATT HOURS



***GW** describes the size of the infrastructure
→ maximum capacity*

***TWh** describes actual consumption
→ 945TWh in 2030 corresponds to an average utilisation
rate of 54%*

DATA CENTERS X ENERGY TRAJECTORIES



Future projections vary:

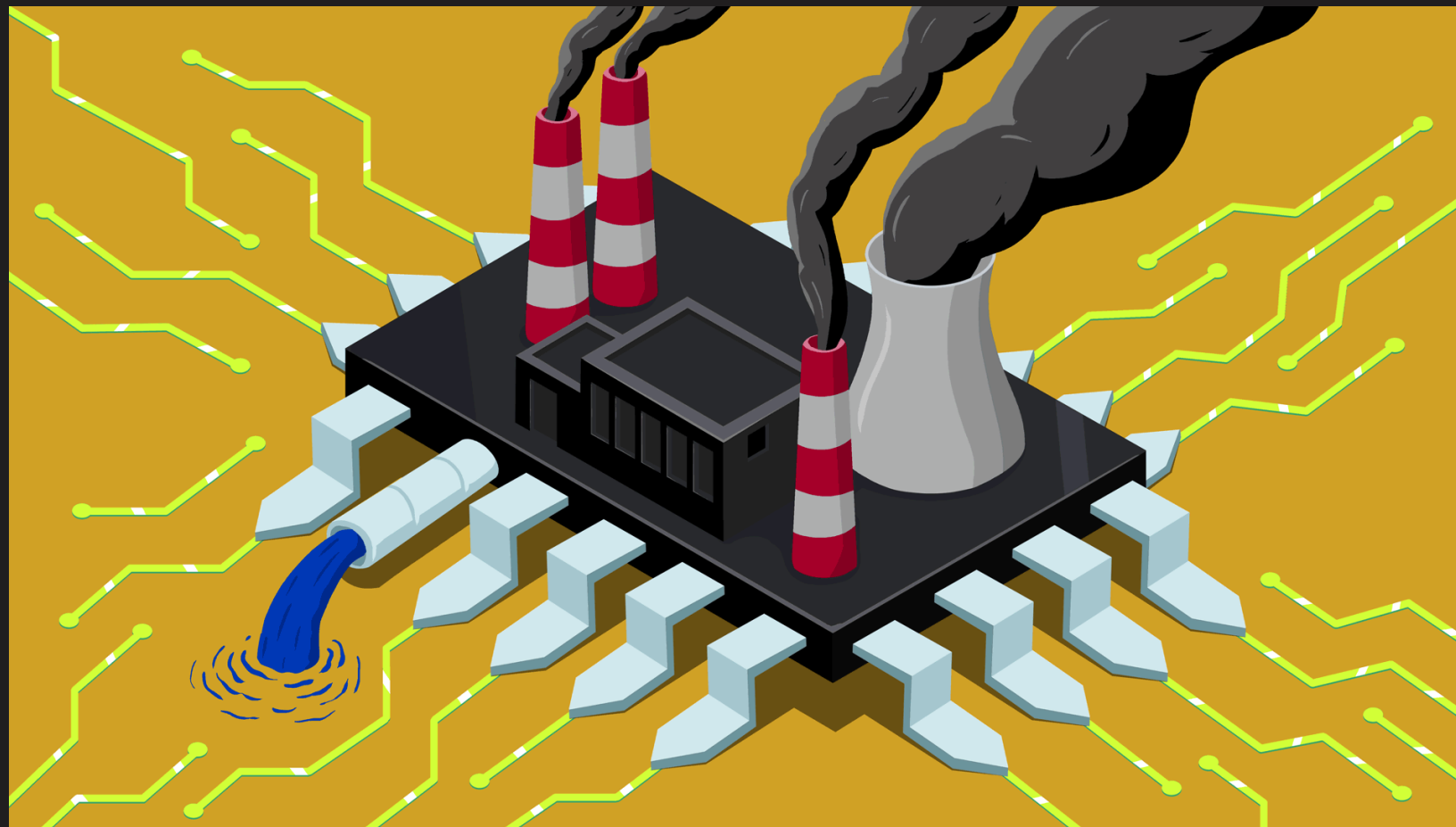
→ 73 TWh in 2026
(IEA, 2024a)

→ 209 TWh by 2030
(Goldman Sachs, 2024)

→ 400 – 500 TWh by 2027-2028
(De Vries, 2023; Semianalysis, 2024).

https://www.umweltbundesamt.de/sites/default/files/medien/11850/publikationen/68_2025_texte_0.pdf

WHY DO AI READY DATA CENTERS USE SO MUCH ENERGY?



<https://fluter.de/energiebedarf-von-KI>

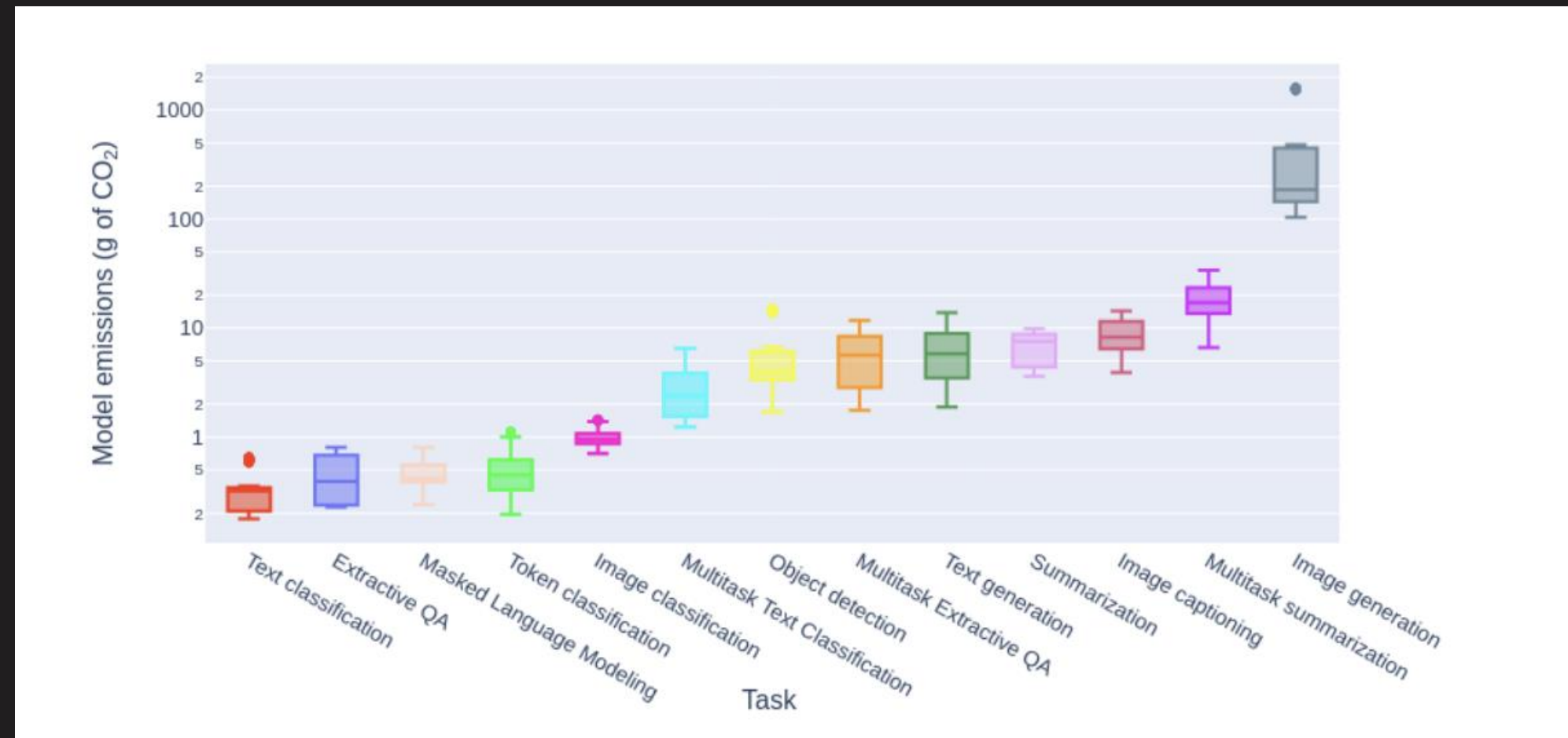
- ✓ parallel, compute-intensive processes
- ✓ GPUs have a much higher energy draw than CPUs:
NVIDIA's Blackwell = 600W
the latest CPUs = 250W
- ✓ multiple GPUs in one server rack, and hundreds of server racks per data center
- ✓ a 2,000 fold increase in the span of a decade



ENERGY CONSUMPTION

Per model basis & inference

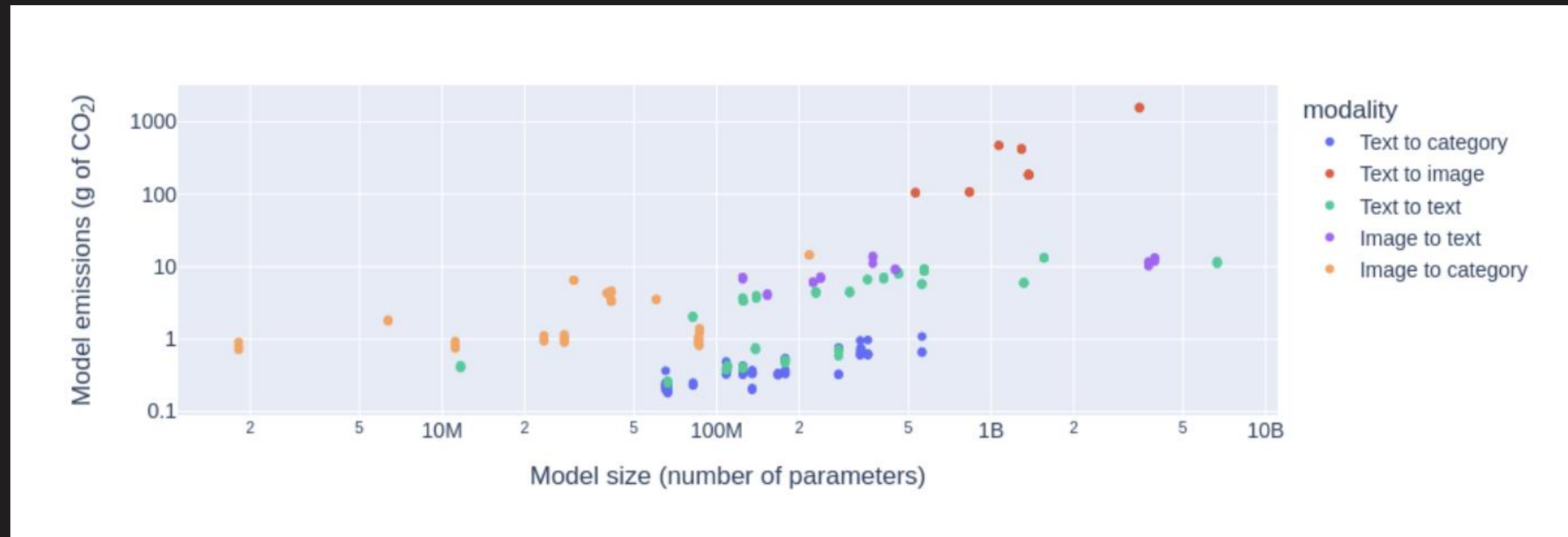
PER TASK ENERGY: CAPABILITY -BASED



Source: <https://dl.acm.org/doi/pdf/10.1145/3630106.3658542>

The tasks examined in and the average quantity of carbon emissions they produced (in g of CO_2eq) for 1,000 queries. The y axis is in logarithmic scale.

PER TASK ENERGY: MODEL SIZE-BASED



Source: <https://dl.acm.org/doi/pdf/10.1145/3630106.3658542>

The 5 modalities examined in the study, with the number of parameters of each model on the x axis and the average amount of carbon emitted for 1000 inferences on the y axis. Both axes are in logarithmic scale.



WHAT CAN WE DO?



PRACTICES OF SUSTAINABLE LLM

Development and Use



SUSTAINABLE USE (INFERENCE) PARADIGMS

Consumer behaviour matters

- **intentionality:** Ask yourself → does this task actually require a large AI model?
- **invisible impact:** leads to thoughtless overuse → unlike a fuel in your car or electricity bill at home, there's no feedback loop

Practices to reduce energy consumption

- Start fresh conversations rather than accumulating long threads → larger context = more compute
- Use task-specific tools for simple jobs (e.g. DeepL for translation, not ChatGPT)
- Only use AI when a search engine, calculator, or a moment of thinking won't suffice

SUSTAINABLE DEVELOPMENT PARADIGMS



Run training in low-carbon electricity regions: A single infrastructure decision that passively reduces emissions across every training run.

Use pre-trained models: Fine-tuning an existing model uses less energy than training from scratch; default to transfer learning unless a novel architecture is strictly required (fine-tune Bert-scale model, Qwen, etc.)

Put a budget on experimentation: Cap trials, enforce compute limits, and replace brute-force hyperparameter practices

Implement Stopping: Don't waste energy training a model that has stopped improving (monitor performance plateaus)

Batch Inferences: Group requests into batches rather than processing them individually to maximise hardware utilisation and cut per-prediction energy cost.

Open-source datasets, evaluation harnesses, model cards, and tokenizers: Sharing reusable assets across the community avoids redundant reproduction of expensive artefacts; impact scales with adoption.



METHODS FOR ESTIMATING ENERGY CONSUMPTION AND CARBON FOOTPRINT OF LLMS

Exercise



EXERCISE 1: ESTIMATING CARBON EMISSIONS AND POWER CONSUMPTION WITH CODECARBON

- ✓ Understand CodeCarbon's methodology for estimating carbon emissions.
- ✓ Learn the relationship between carbon intensity (C) and energy consumption (E).
- ✓ Implement a simple heuristic to estimate carbon emissions using GPU specifications.
- ✓ Compare heuristic estimates with actual CodeCarbon measurements and analyze discrepancies.



EXERCISE



HOW DATA CENTER LOCATION (AND THE ENERGY MIX) AFFECTS ENVIRONMENTAL IMPACT

WHERE IS THE ENERGY COMING FROM?



IEA: globally 60% of the data center energy is generated from **fossil fuels** (coal and natural gas).

In the US alone, data centers are estimated to generate more than 105 million tons of CO₂eq = 2% of US emissions in 2023

(🚗 ≈ annual emission of 23 mil. cars)

→ *This estimate represents the emissions from the energy that data centers physically consume (i.e. **location-based emissions**), rather than any contract that data center operators have to buy renewable energy credits (i.e. **market-based emissions**), which are often reported by the operators.*

CAN DATA CENTERS BE FULLY POWERED BY RENEWABLE ENERGY?



the amount of electricity required by data centers can **swing massively** from one moment to another:

→ **intense but fluctuating pattern** that uses around 80% of capacity for training and 40-60% for inference

While fossil fuel generators can be very responsive to fluctuations in demand, renewable energy sources - and nuclear power plants - cannot

→ hard to control the output of solar panels or wind turbines, which will respond to changes in the sun's rays or wind speed (while nuclear power plants always have a near-steady output).

→ theoretically possible to compensate for this with equipment

→ batteries or condensers, this can add significant costs, making these options much **more expensive for builders and operators**



WILL NUCLEAR POWER SOLVE THE DATA CENTER ENERGY CRISIS?



[CC BY-NC-ND](#)

While new reactors are continuing to be built globally, this takes, on average, **6 to 8 years**;

when existing decommissioned nuclear reactors are put back online (e.g. Three Mile Island and Duane Arnold Energy Center, which are set to be revived by Microsoft and Google), this **can take 3-5 years**.

in countries like France, where surplus nuclear energy is readily available, connecting new data centers to the existing electricity grid can take **up to 5 years** due to the permitting and construction procedures required.

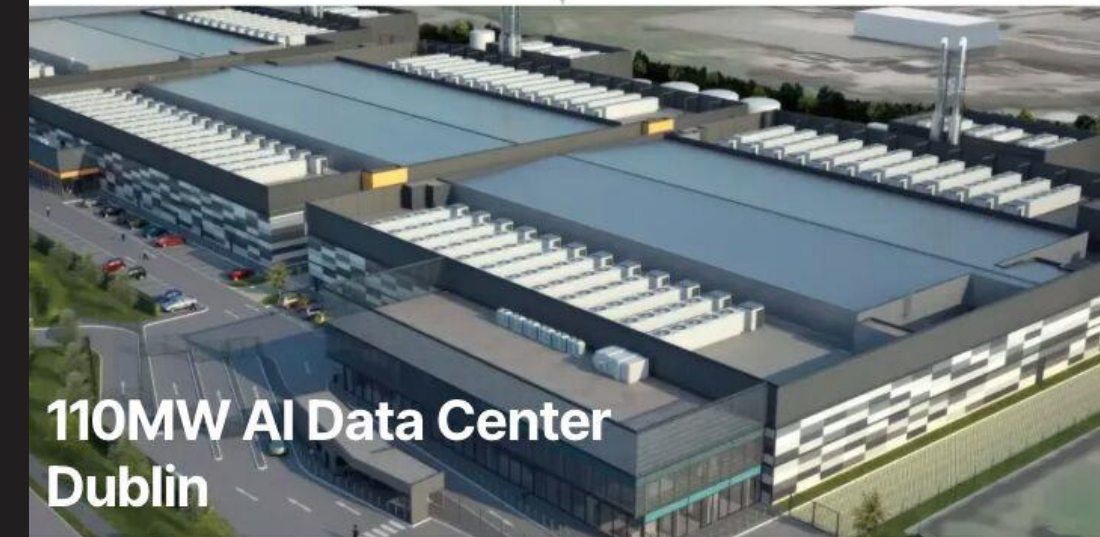
BYPASS THE GRID CONNECTION QUEUE

- **xAI (Memphis):** trucked in gas turbines to power 100,000 GPUs; bypassed the grid entirely [🔗](#)
- **OpenAI (Texas):** ordered 29 gas turbines for its Stargate data centre
Some developers are bolting retired aircraft jet engines onto trailers as makeshift power plants [🔗](#)
- **Australia:** data centre emissions up 16% in a single year; one developer approved a gas plant and refused to explain why it didn't choose renewables [🔗](#)
- **AI Power Demand in the US** to grow from ~3GW in 2023 to over 28GW by 2026 [🔗](#)
- Most of these behind-the-meter gas projects were announced in **2025 alone**

100% gas, off-grid Data Center in Ireland goes live



Rolls Royce (mtu) 4000 Series RICE (natural gas)



110MW AI Data Center
Dublin

+1.1%

Ireland Emissions Increase
569,572 tCO₂e

[image source](#) [🔗](#)



LOCATION MATTERS: FINDING GREEN PLACES TO TRAIN LLMS

Exercise

EXERCISE GOALS



- ✓ Understand how location affects carbon emissions through regional carbon intensity differences.
- ✓ Analyze global carbon intensity data to identify optimal training locations.
- ✓ Calculate estimated emissions for the same workload across different countries.
- ✓ Visualize the environmental impact of location choices using charts and maps.



EXERCISE



HOW TO IMPLEMENT A CARBON-TRACKING SYSTEM IN YOUR AI DEVELOPMENT WORKFLOW

Exercise



EXERCISE GOALS

- ✓ Learn how to import and initialize CodeCarbon's EmissionsTracker in a Python training script.
- ✓ Understand where to place the tracker's start() and stop() methods in a distributed training loop.
- ✓ Configure CodeCarbon to output emission reports.
- ✓ Run a training job with carbon tracking enabled and analyze the generated emission report.



EXERCISE



BREAK!



BEYOND CARBON: OTHER DIMENSIONS OF SUSTAINABILITY IN AI DEVELOPMENT

'Carbon tunnel vision'











**No data on other
life cycle stages**

AI DEVELOPMENT



Raw materials
Chile



Raw materials
Dem. Rep. Congo



Production
Taiwan



Disposal
Ghana

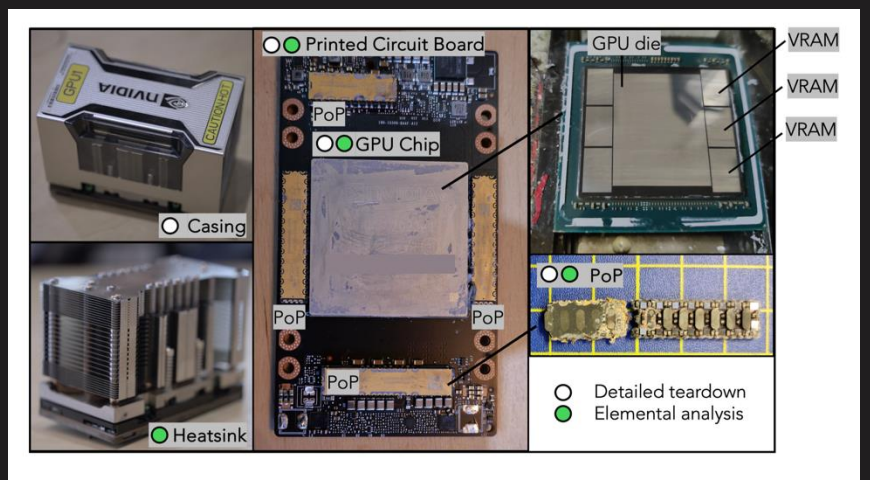
RESEARCH GOAL



01 MORE-THAN-CARBON ASSESSMENT

02 OPEN-SOURCE PRIMARY DATA

METHOD



Step 1 GPU disassembly

Analysis performed by: Jérôme SUDRES
Reported by: Jérôme SUDRES
Date: 2023/03/01

Résultats d'analyse de Hublot - Main D3

Notes: Cette feuille présente les résultats d'analyse de la composition chimique de Hublot - Main D3. Afin de garantir la fiabilité de cette analyse, le matériel obtenu a été caractérisé à l'aide de :

Composition chimique des échantillons :

Éléments	Teneur (Wt%)	2.4 (Wt%)	Teneur (g T-1)
Ag	0.05	0.01	511
Al	1.89	0.14	18813
As	0.00	0.00	11
Br	0.01	0.00	51
B	0.27	0.00	2715
Ba	0.19	0.01	1176
Bi	0.10	0.01	1005
Ca	1.10	0.04	29053
Co	0.00	0.00	1
Cr	0.00	0.00	1
Cu	0.00	0.00	1
Fe	0.37	0.01	3681
Ge	0.01	0.00	79
Ir	0.00	0.00	1
Mg	0.12	0.01	1152
Mn	0.04	0.00	417
Mo	0.02	0.00	189
Nb	0.07	0.00	705
Ni	0.17	0.01	3499
Pb	0.02	0.00	213
Pd	0.00	0.00	41
Re	0.00	0.00	1
Rh	0.00	0.00	1
S	16.16	7.10	161107
Se	0.03	0.02	98773
Si	0.23	0.02	2306
Ti	0.00	0.00	11
V	0.00	0.00	11
Zn	0.11	0.00	1087
Indétecté	16.67		166726
Pourcentage	73.86	0	239813
TOTAL	98.23		982340

Informations complémentaires :

Le résultat d'analyse en ICP-OES est la somme de l'analyse chimique. En conséquence, la somme des résultats de la réglementation des éléments des échantillons doit être égale à 100%. Les échantillons analysés ont été analysés par la réglementation des échantillons, les 1.28 % manquants au 100%, provenant de ces éléments non analysés à l'ICP-OES.

Informations d'analyse :

Présentation d'analyse : Le résultat d'analyse est la somme de l'analyse chimique. En conséquence, la somme des résultats de la réglementation des échantillons doit être égale à 100%. Les échantillons analysés ont été analysés par la réglementation des échantillons, les 1.28 % manquants au 100%, provenant de ces éléments non analysés à l'ICP-OES.

Nombre d'échantillons analysés :

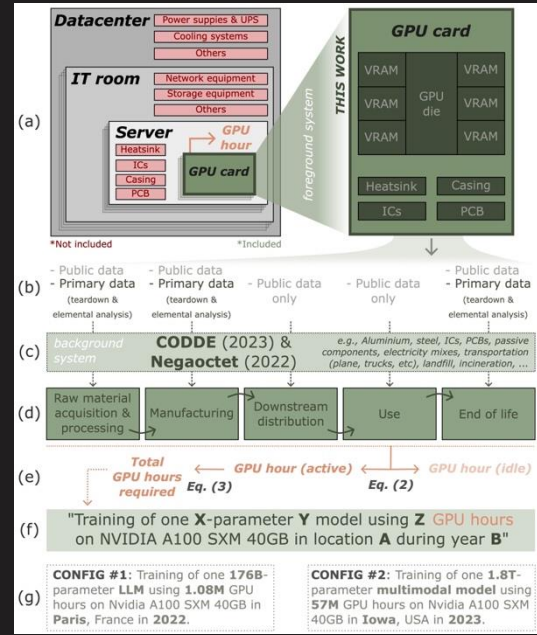
Nombre d'analyse de l'ICP :

100%

100%

100%

Step 2 elemental analysis



Step 3 LCA

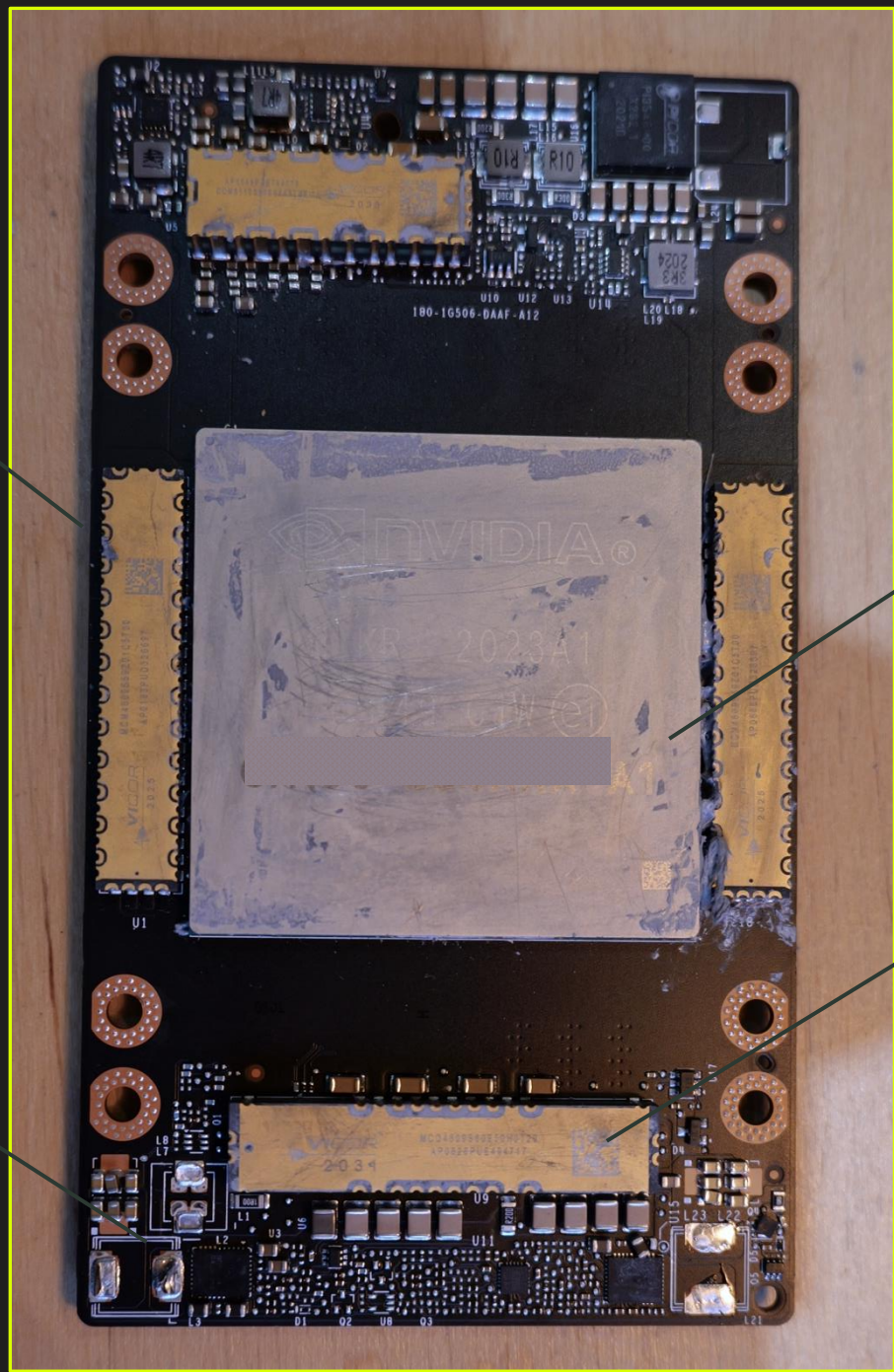
STEP 1: NVIDIA A100 SXM 40GB



Casing ○



Printed Circuit Board ○ ●



GPU chip ○ ●

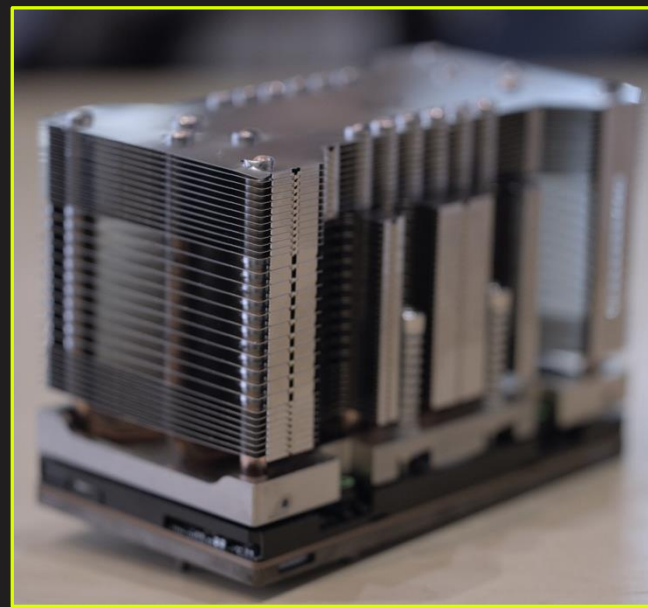


VRAM

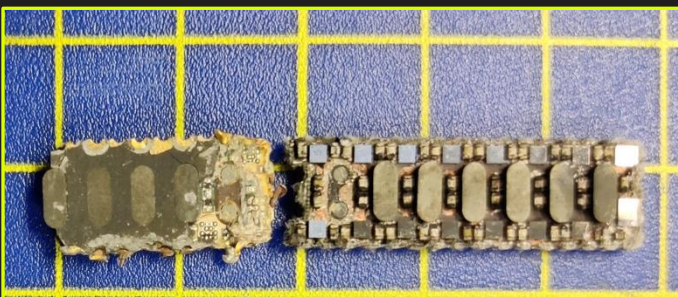
VRAM

VRAM

Heatsink ●



Power-on-Package ○ ●



○ Detailed teardown

● Elemental analysis

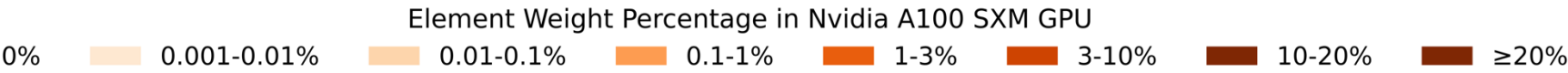
POLYGLOT

STEP 2: INDUCTIVELY COUPLED PLASMA OPTICAL EMISSION SPECTROSCOPY (ICP-OES)

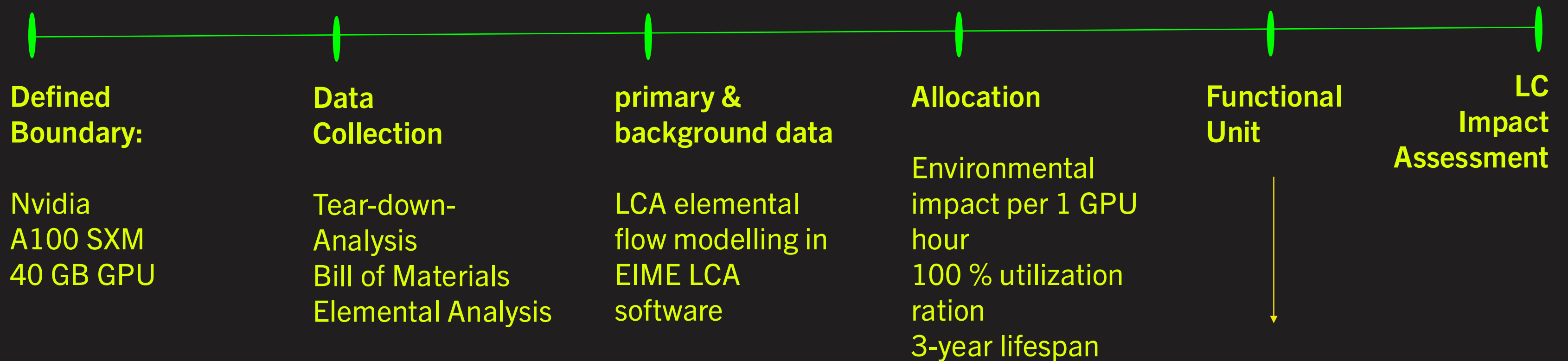


Element	Abk.	Gesamt
Silber	Ag	0,553
Aluminium	Al	6,90
Arsen (*)	As	0,034
Gold	Au	0,043
Bor	B	0,782
Barium	Ba	4,05
Beryllium (*)	Be	0,024
Bismut	Bi	2,80
Calcium	Ca	5,78
Cadmium (*)	Cd	0,004
Cobalt (*)	Co	0,017
Chrom (*)	Cr	5,66
Kupfer (*)	Cu	1370
Eisen	Fe	45,5
Kalium	K	0,032
Lithium	Li	0,073
Magnesium	Mg	0,311
Mangan	Mn	0,918
Natrium	Na	0,691
Nickel (*)	Ni	11,1
Blei (*)	Pb	0,656
Palladium	Pd	0,200
Platin	Pt	0,057
Antimon (*)	Sb	0,085
Selen	Se	0,094
Silizium	Si	13,4
Zinn	Sn	20,3
Titan	Ti	1,49
Thallium (*)	Tl	0,023
Vanadium	V	0,012
Zink (*)	Zn	1,18
Strontium	Sr	0,005

1 H Hydrogen																	2 He Helium
3 Li Lithium	4 Be Beryllium											5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium											13 Al Aluminum	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson
			58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium	
			90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium	



STEP 3: LCA ISO 14040:2006/ ISO 14044:2006 | PEF EF 3.0 V2.0 (MIDPOINT)



The complete training of one **1.8T-parameter GPT-4** model using **57M GPU hours** on Nvidia A100 in SXM 40GB in Iowa, USA during **2023**, resulting in a production-ready model capable of providing a inference as a service.

9,700,000



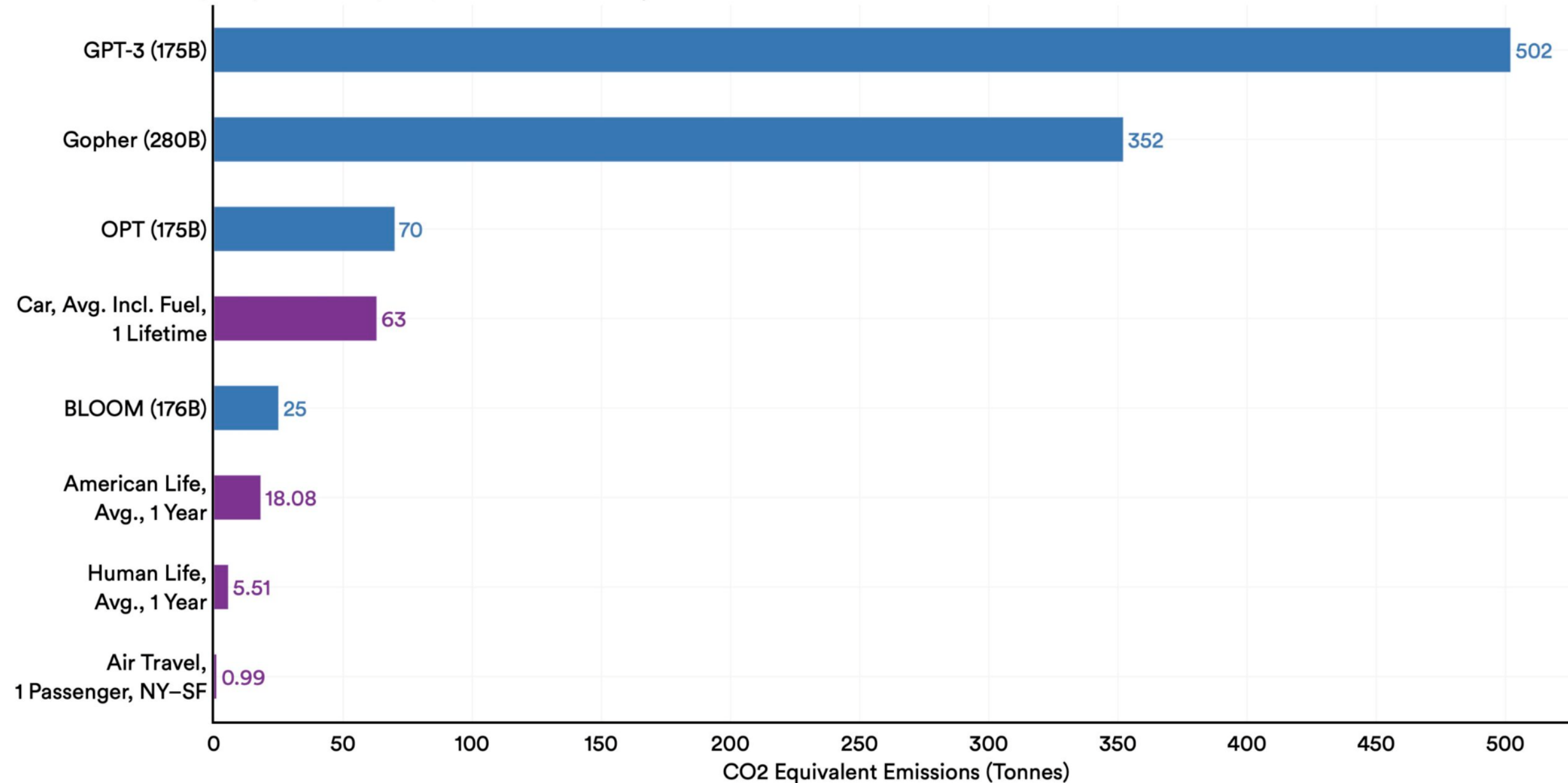
$CO_{2\text{ eq.}}$ *Emissions of Training GPT-4*

GPT-4 // 9,700 TONNES CO₂ EQ.



CO2 Equivalent Emissions (Tonnes) by Selected Machine Learning Models and Real Life Examples, 2022

Source: Luccioni et al., 2022; Strubell et al., 2019 | Chart: 2023 AI Index Report

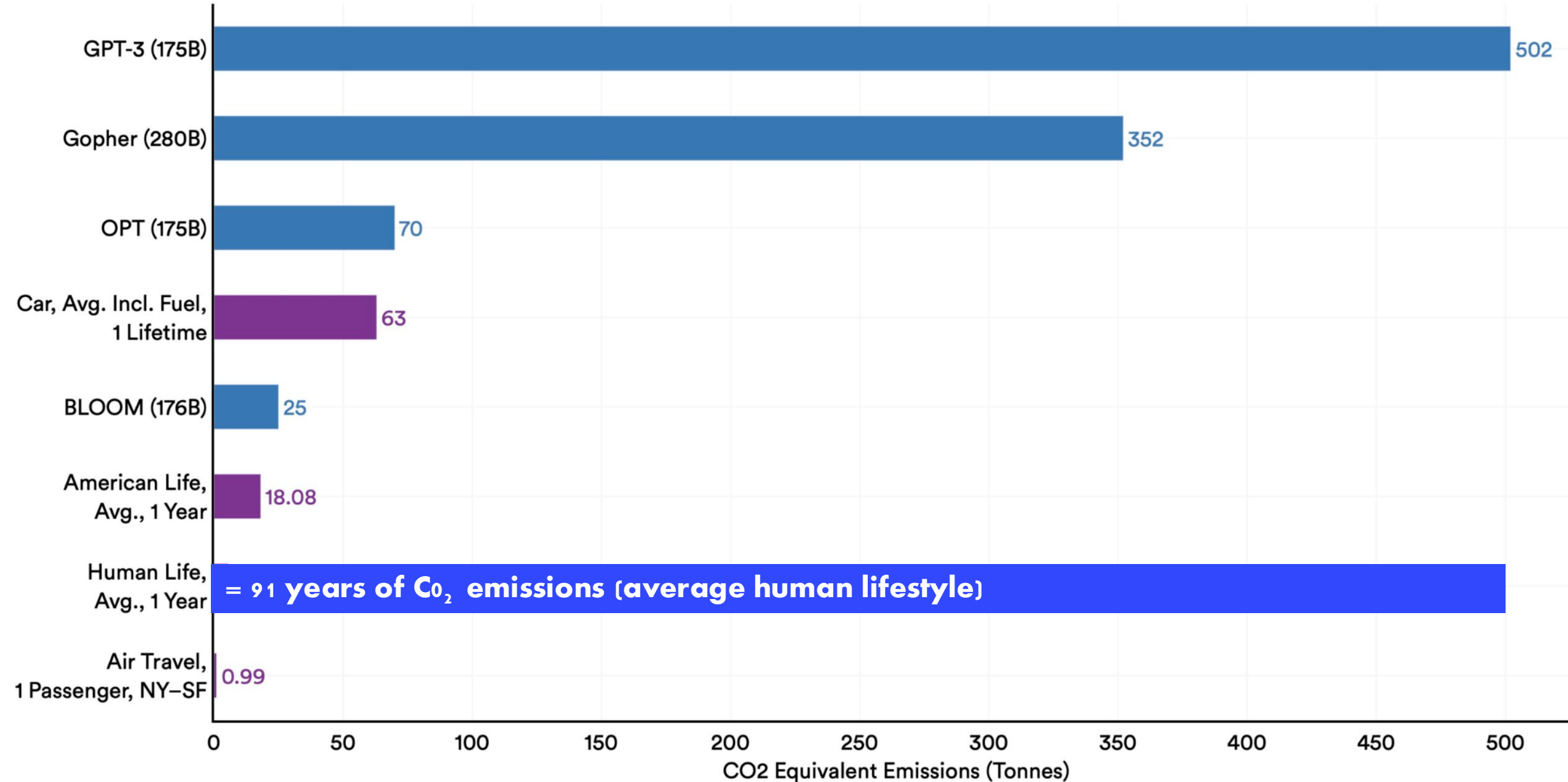


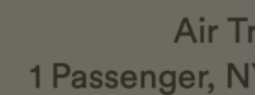
GPT-4 // 9,700 TONNES CO₂ EQ.



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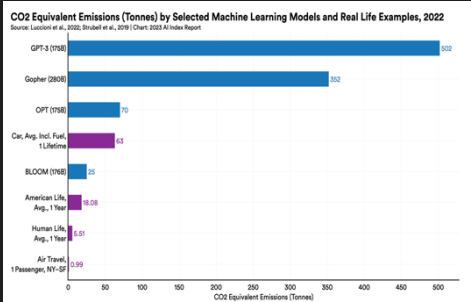


500

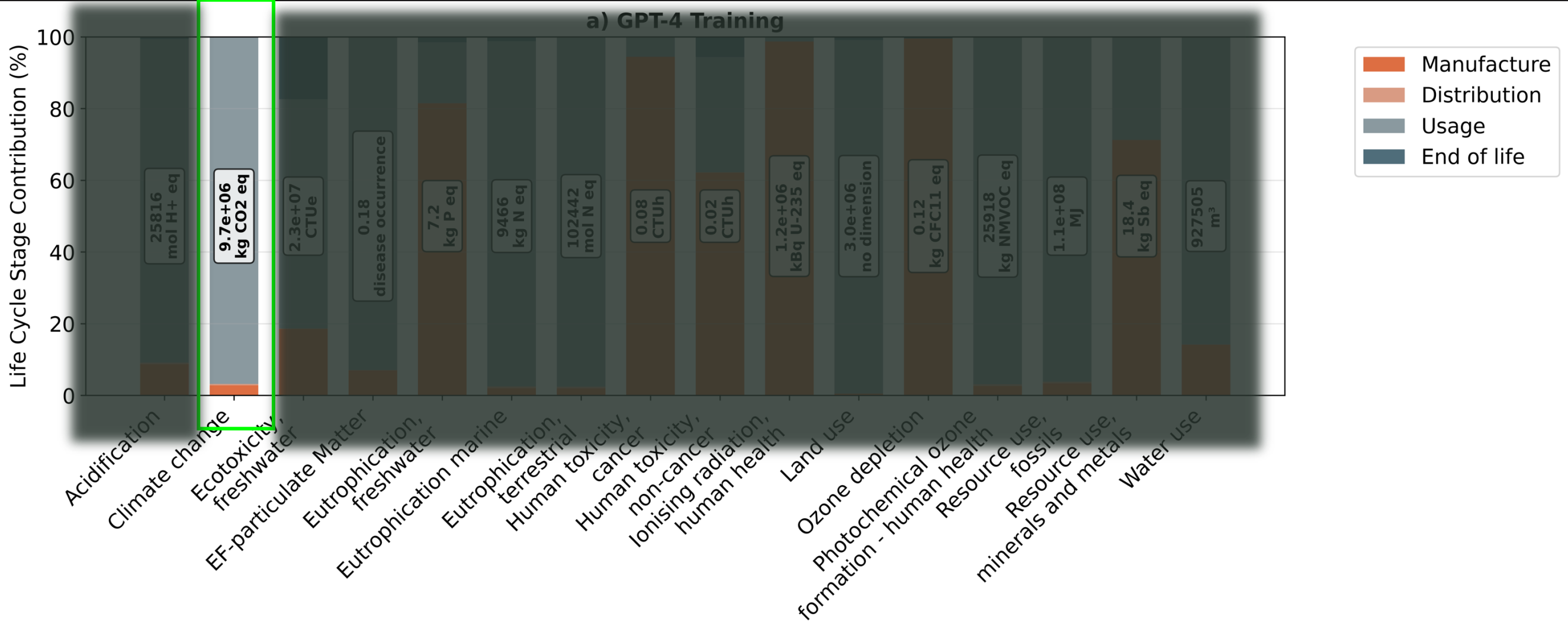
GPT-4 // 9,700 TONNES CO₂ EQ.



CO₂ eq. Emissions of Training GPT-4



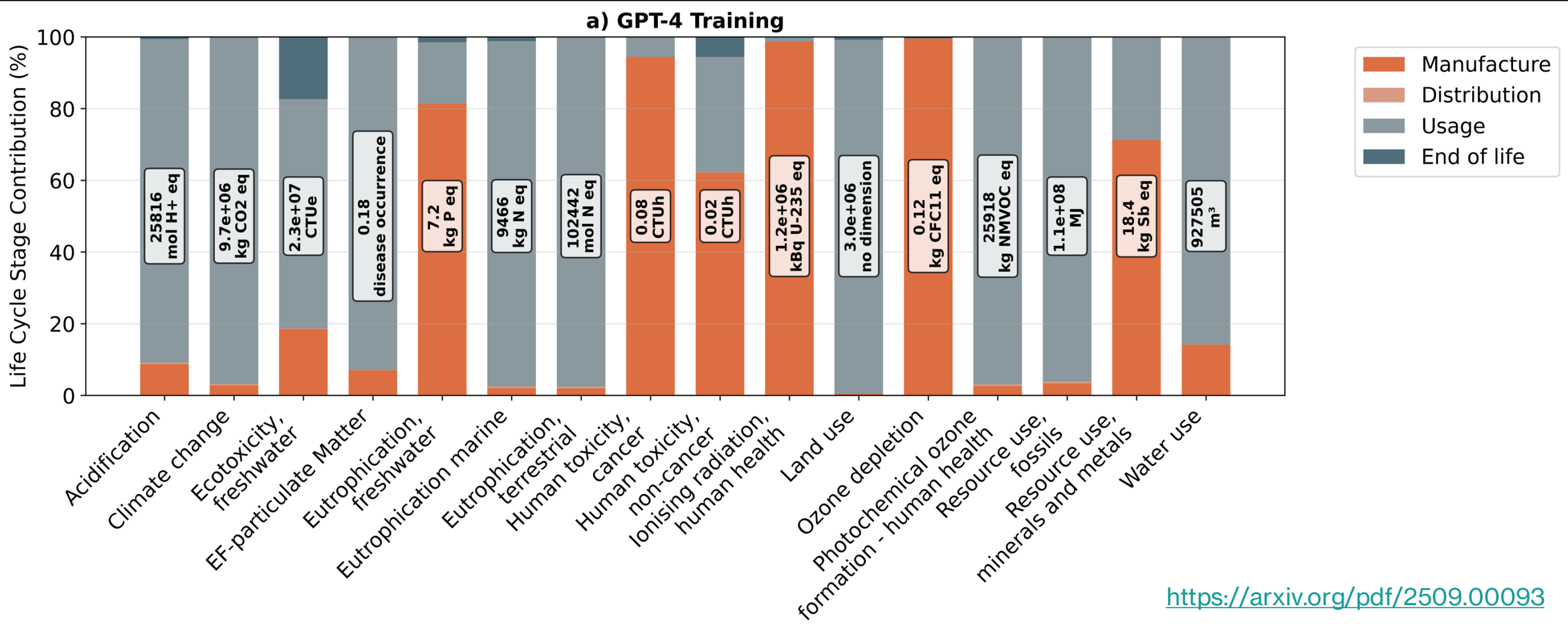
ENVIRONMENTAL IMPACTS OF TRAINING GPT-4



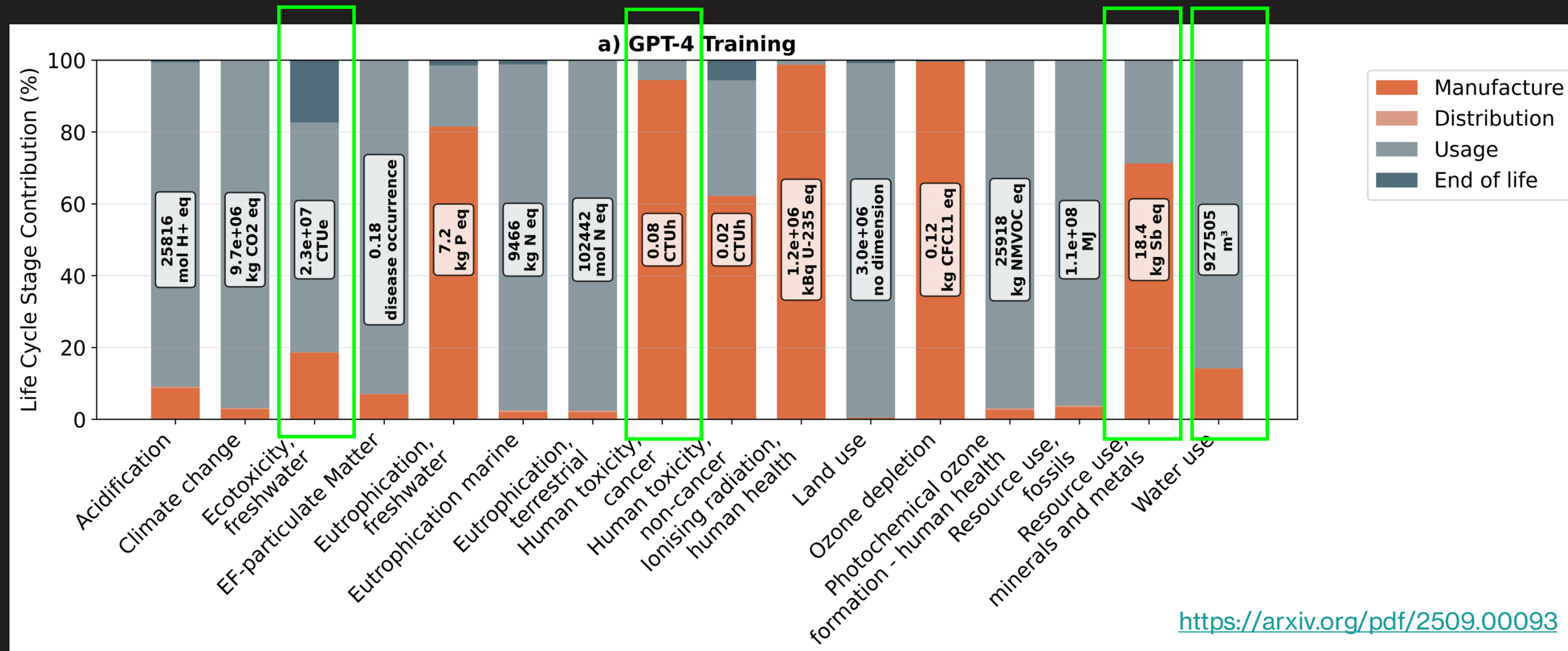
The use phase accounts for 96.8% of carbon emissions.

3.2% - 300 tons CO2

ENVIRONMENTAL IMPACTS OF TRAINING GPT-4



ENVIRONMENTAL IMPACTS OF TRAINING GPT-4



ENVIRONMENTAL IMPACTS OF TRAINING GPT-4



23,000,000 CTUe | Ecotoxicity (freshwater)

Comparative Toxic Units for ecosystems

toxicity score ,*how harmful the product could be to plants, animals, and environmental systems*’ – significant ecological impact

0.08 CTUh | Human toxicity (cancer)

Comparative Toxic Units for humans

does not convey the scale or geographic distribution of health outcomes – could affect one person severely or one million people minimally

18.4 kg SB eq. | Resource use (minerals and metals)

Antimony equivalent

Measuring resource scarcity based on availability, extraction difficulty, and depletion rate of the scarcest element we have

927.5 million liters | Water use

Daily water supply for about 12,700 people for an entire year (200 liter/day)



<https://arxiv.org/pdf/2509.00093>

ENVIRONMENTAL IMPACTS OF TRAINING GPT-4



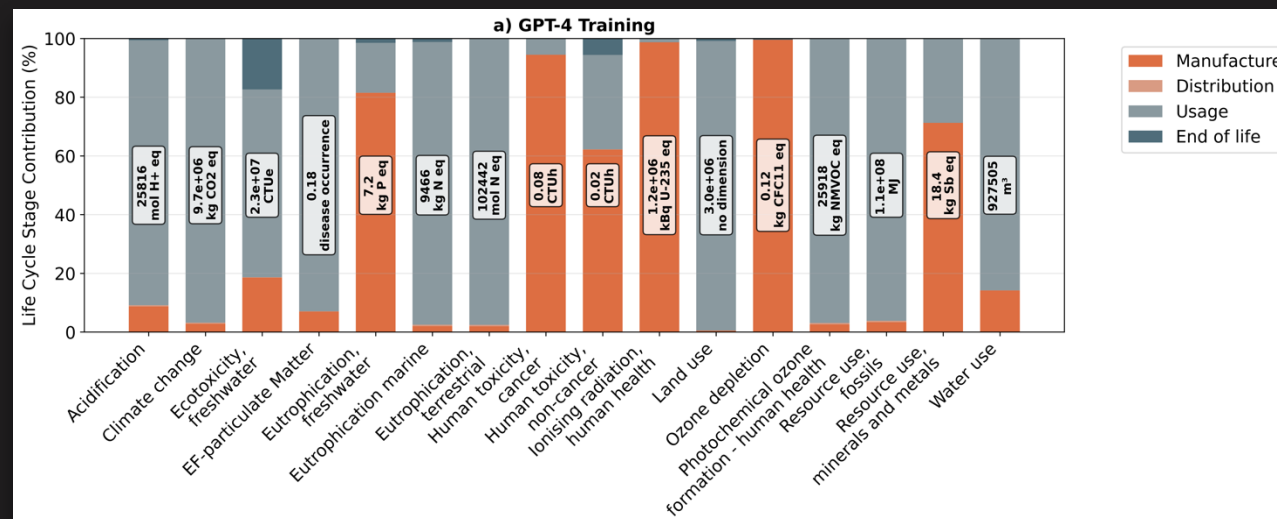
SUMMARY

01 Manufacturing: material-related and toxicity impact categories

02 Distribution: overall relatively modest contribution

03 Use phase: climate change related impact categories

04 End-of-Life: ecotoxicity and land use categories | Data limitations



<https://arxiv.org/pdf/2509.00093>

GPU EMBEDDED IMPACTS CALCULATOR



This calculator estimates the embedded impacts (manufacturing, distribution, end-of-life) of a graphics card based on 16 Product Environmental Footprint (PEF) impact criteria (excluding the use phase):

 <https://data.hubblo.org/gpu>



NOW WHAT..?

Now we got the numbers – what are the next steps?

01 Using them

02 understand the **implications** of the units and metrics

03 popularize the use of “more than carbon” units and metrics

KEY TAKE-AWAYS



01 carbon-only accounting misrepresents the globally distributed impact profile, creating a dangerous blind spot in the sustainability discourse.

02 we need multi-criteria environmental assessment frameworks that account for geographic, temporal, and impact-specific distributions.
(for that we need more transparency from the industry).

03 regulation must address the geographical disconnect between AI access and environmental burdens.

Let's not only talk about carbon emissions.



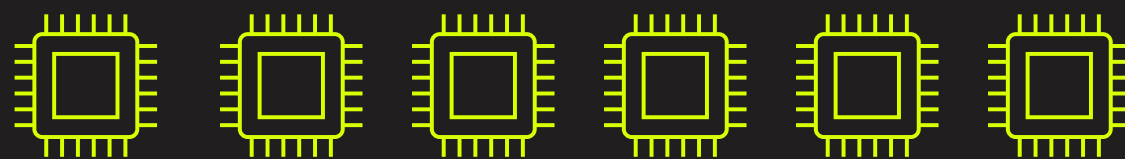
BEYOND CARBON: THE MATERIAL CONSUMPTION OF LLM DEVELOPMENT



FROM FLOPS TO FOOTPRINTS



Compute per GPU



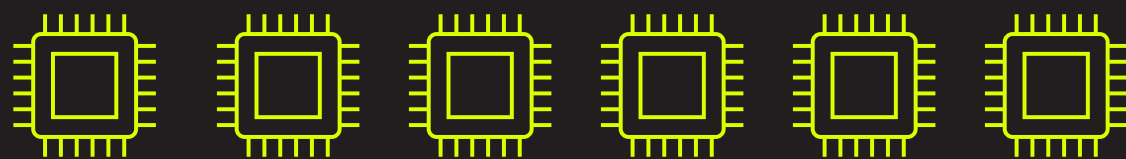
Compute Budget per AI model



FROM FLOPS TO FOOTPRINTS



Compute per GPU



Annual Computational Throughput (100% utilization)

$$\begin{aligned} &= (312 \times 1012) \times (365 \times 24 \times 60 \times 60) \\ &= 9.8 \times 10^{21} \text{ FLOPs} \end{aligned}$$

(1)



Compute Budget per AI model

$$\text{Compute Budget} = 6 \times N \times D$$

(2)

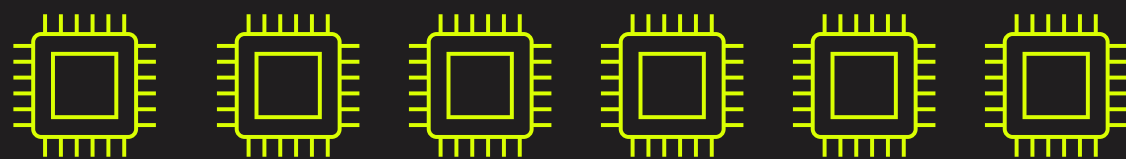
Where N is the number of model parameters and D the number of training tokens.



FROM FLOPS TO FOOTPRINTS



Compute per GPU

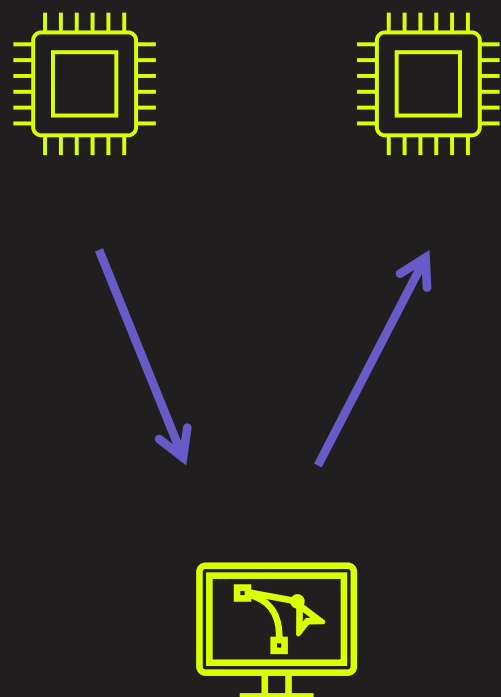


Compute Budget per AI model

$$\text{Required GPUs} = \frac{\text{Eq. (2)}}{\text{Eq (1) x lifespan}} = \frac{\text{Compute Budget}}{\text{Annual Computational Throughput} \times \text{Lifespan}} \quad (3)$$



FROM FLOPS TO FOOTPRINTS



$$\text{Required GPUs} = \frac{\text{Eq. (2)}}{\text{Eq (1) x lifespan}} = \frac{\text{Compute Budget}}{\text{Annual Computational Throughput} \times \text{Lifespan}} \quad (3)$$

Adding additional variables:

✓ Training efficiency

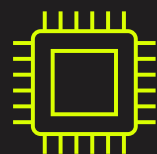
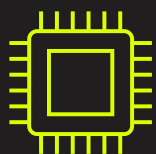
$$\text{Scaling factor} = 1 \text{ MFU where MFU} \in [0.20, 0.60] \quad (4)$$

✓ The adjusted GPU count is then computed as:

$$\text{GPU adjusted} = \text{GPU required} \times \text{Scaling factor} \quad (5)$$



FROM FLOPS TO FOOTPRINTS



$$M_{\text{element}} = m_{\text{element}} \times \text{GPU adjusted}(n) \quad (6)$$

Where m_{element} is the mass of a specific element detected in the analyzed GPU, and $\text{GPU adjusted}(n)$ corresponds to the adjusted GPU requirement accounting for real-world MFU and hardware lifespan.



EQUATION OVERVIEW

$$\text{Annual Computational Throughput} = (312 \times 10^{12}) \times (365 \times 24 \times 60 \times 60) = 9.8 \times 10^{21} \text{ FLOPs} \quad (1)$$

Following Equation 1; at 100% utilization over one year, this corresponds to 9.8×10^{21} FLOPs annually.

$$\text{Compute Budget} = 6 \times N \times D \quad (2)$$

Where N is the number of model parameters and D the number of training tokens.

$$\text{Required GPUs} = \text{Compute Budget} / \text{Annual Computational Throughput} \times \text{Lifespan} \quad (3)$$

$$\text{Scaling factor} = 1 \text{ MFU where } \text{MFU} \in [0.20, 0.60] \quad (4)$$

$$\text{The adjusted GPU count is then computed as: GPU adjusted} = \text{GPU required} \times \text{Scaling factor} \quad (5)$$

$$M_{\text{element}} = m_{\text{element}} \times \text{GPU adjusted}(n) \quad (6)$$

Where m_{element} is the mass of a specific element detected in the analyzed GPU, and $\text{GPU adjusted}(n)$ corresponds to the adjusted GPU requirement accounting for real-world MFU and hardware lifespan.



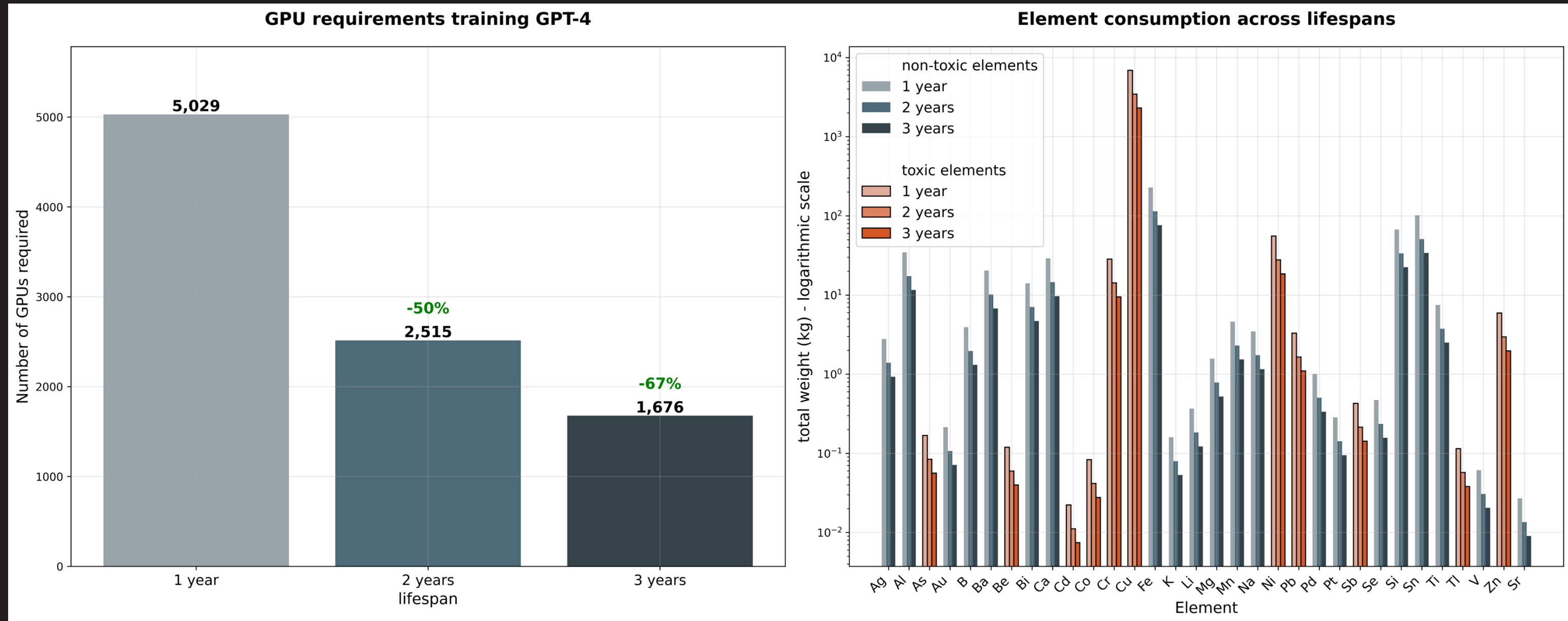
GPU REQUIREMENT PER MODEL TRAINING

Table 4. Total GPU requirements for training selected AI models under varying hardware lifespan (1–3 years) and model FLOPs utilization (MFU) scenarios (20% lower bound, 50% upper bound). Values are rounded up to the nearest integer.

Model	FLOPs	Lifespan = 1		Lifespan = 2		Lifespan = 3	
		MFU 20%	MFU 50%	MFU 20%	MFU 50%	MFU 20%	MFU 50%
GPT-4	1.73×10^{25}	8,800	3,520	4,400	1,760	2,934	1,174
Amazon Titan	4.8×10^{24}	2,439	976	1,220	488	814	326
Mistral Large 2	1.48×10^{24}	751	301	377	151	251	151
LLama 2	8.4×10^{23}	427	171	214	85.38	143	57
DeepSeekLLM	8.04×10^{23}	409	164	205	82	137	55
BLOOM	3.86×10^{23}	196	78	99	40	66	27
GPT-3.5	3.15×10^{23}	160	64	80	32	54	22
Falcon	2.4×10^{23}	122	49	61	25	41	17
Pythia	2.16×10^{23}	11	4	6	3	4	2

<https://arxiv.org/pdf/2512.04142>

RESOURCE CONSUMPTION TRAINING GPT-4



<https://arxiv.org/pdf/2512.04142>

→ Hardware lifespan matters



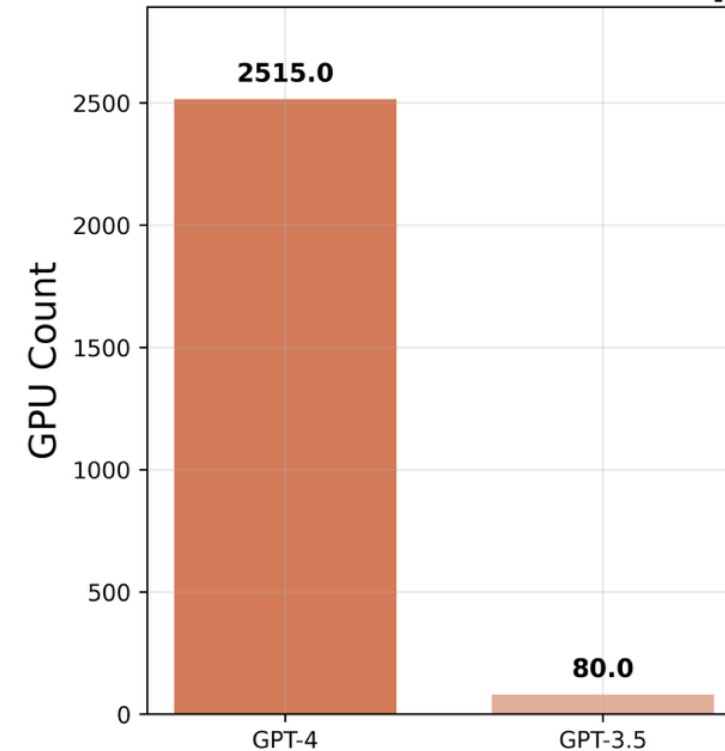
PERFORMANCE VS RESOURCE CONSUMPTION

PERFORMANCE VS RESOURCE CONSUMPTION

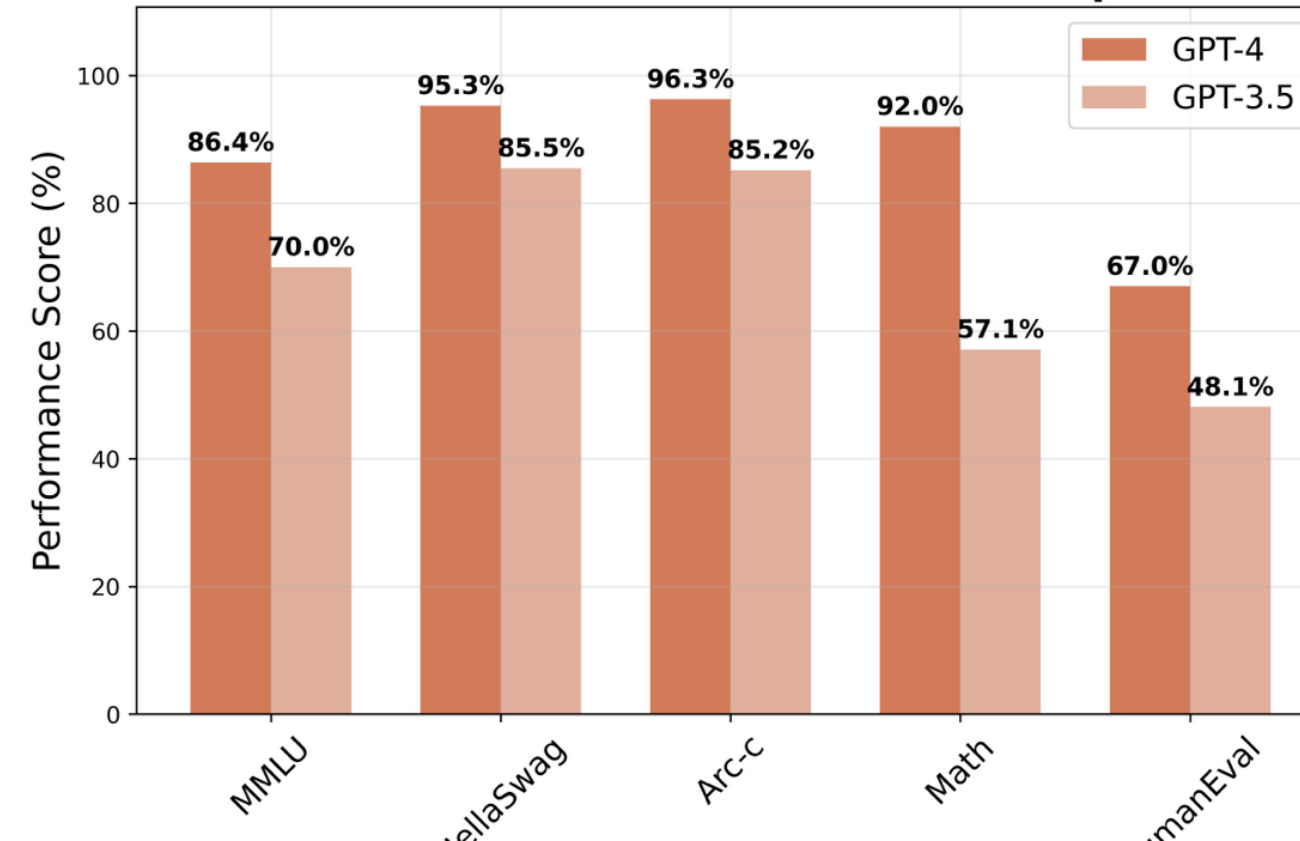


Model Performance and Resource Comparisons

a) GPT-4 vs GPT-3.5: Resource Requirements



a) GPT-4 vs GPT-3.5: Performance Comparison



Source: <https://arxiv.org/pdf/2512.04142>

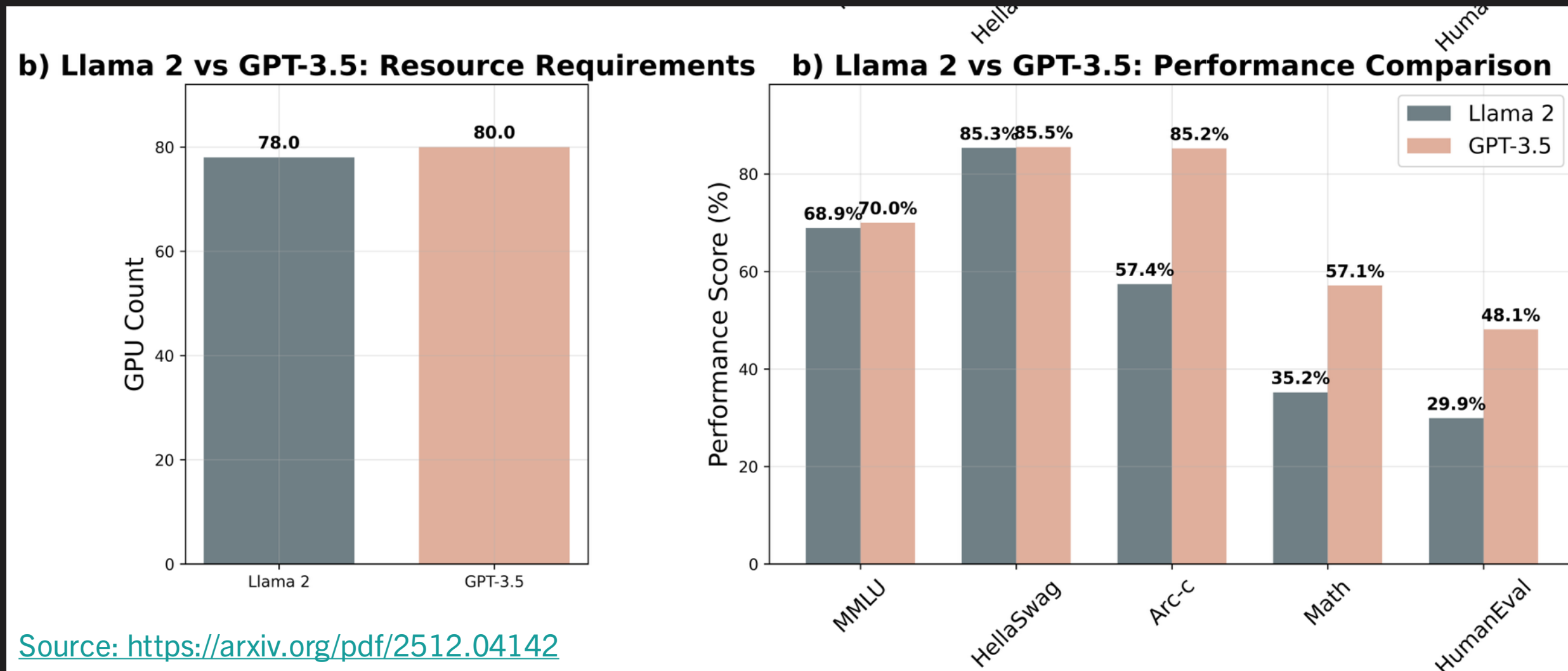
GPT-3.5: 175B Parameter // 300 B Token // MFU: 20% (dense Transformer)

GPT-4: 1.76T Parameter // 222B active Parameter // 13T Token // MFU: 35% (MoE)

→ suggest diminishing returns in terms of performance relative to computational investment.



PERFORMANCE VS RESOURCE CONSUMPTION

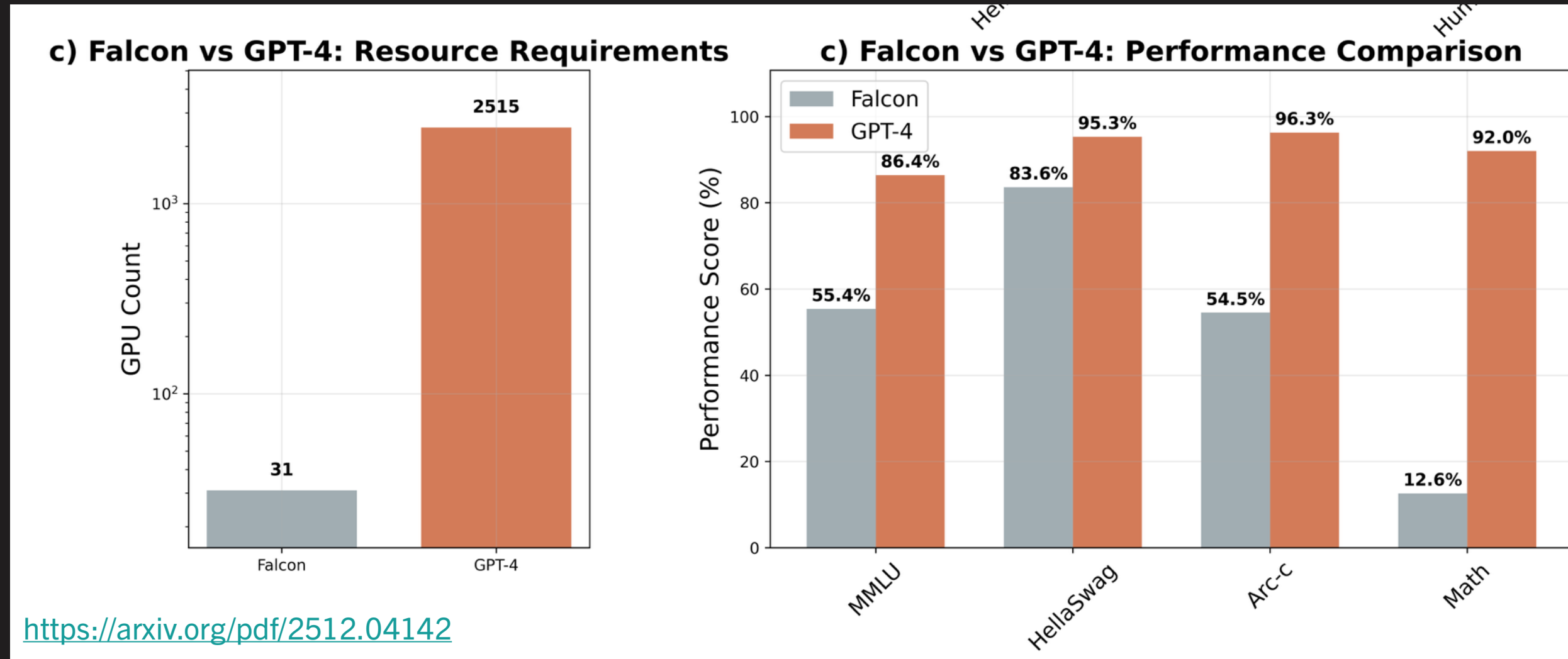


GPT-3.5: 175B Parameter // 300 B Token // MFU: 20%

Llama 2: 70B Parameter // 2T Token // MFU: 53%

→ Similar computational budget/training resources → FLOPs: 10^{23}

PERFORMANCE VS RESOURCE CONSUMPTION



GPT-4: 1.76T Parameter // 222B active Parameter // 13T Token // MFU: 35% (MoE)

Falcon: 40B Parameter // 1 T Token // MFU: 40%

→ *Math benchmark particularly resource intensive capability*

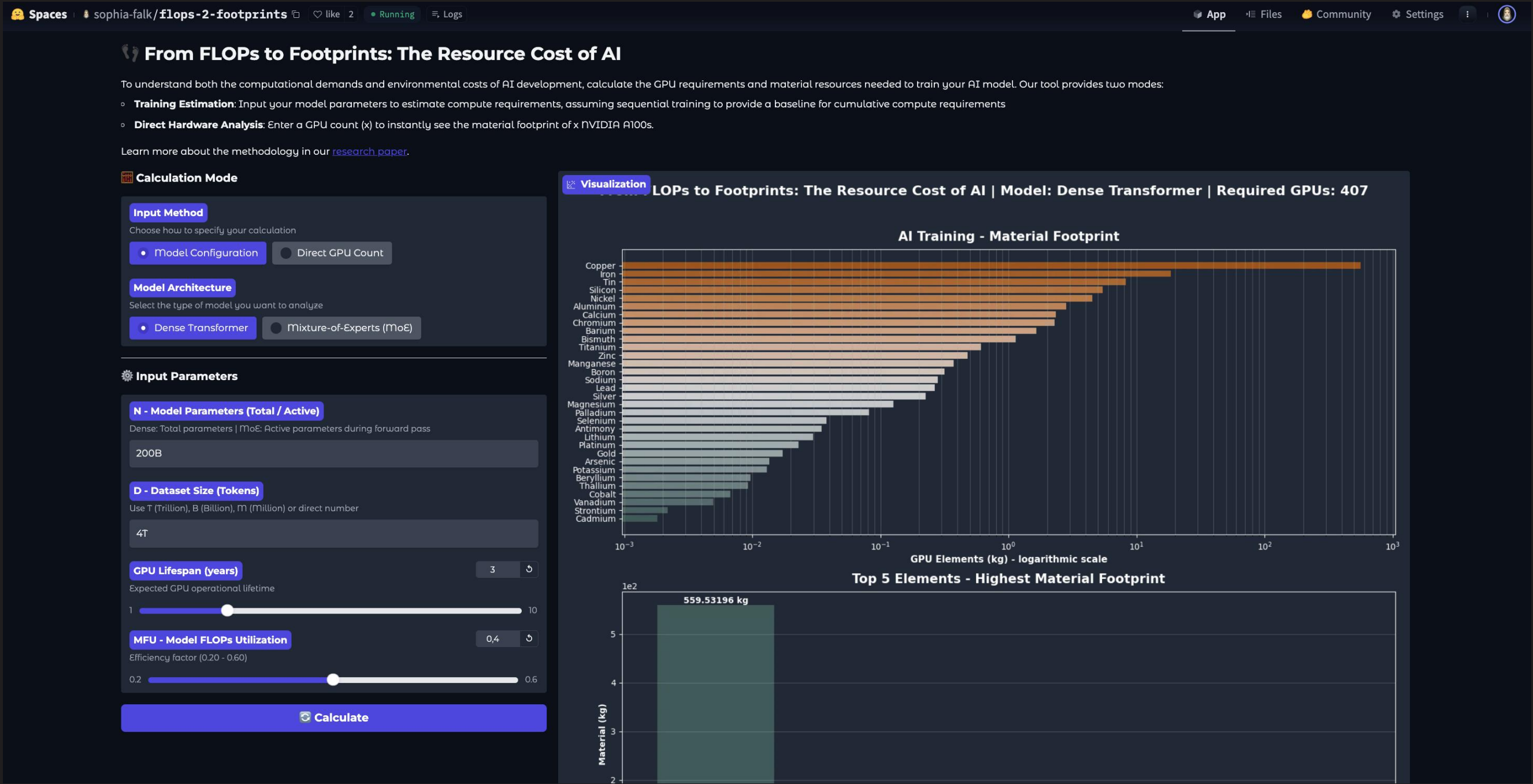


TAKE AWAYS

01 Architectural and algorithmic innovations could offer more sustainable ways to improve model performance than simply scaling models to force intelligence with brute force.

02 We need to include material discussions in discussions about the sustainability and scalability of AI.

TRY THE TOOL TO ASSESS YOUR OWN MODEL



<https://huggingface.co/spaces/sophia-falk/flops-2-footprints>



EXAMPLE: TUCANO 2

the aggregated material footprint of all training runs in the **Tucano 2** series (60% MFU // 3 years hardware lifespan):

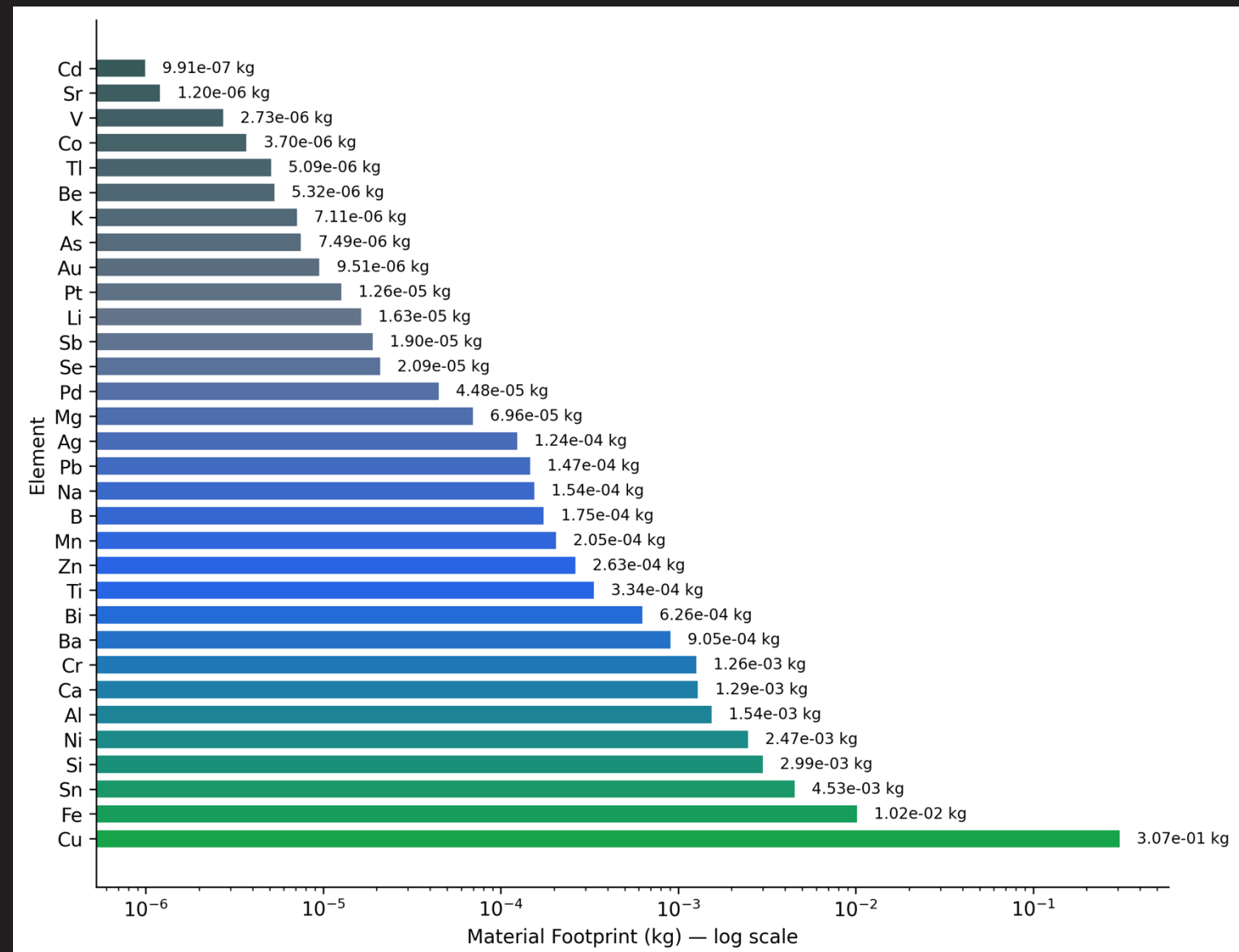
- 0.307 kg of copper
- 0.010 kg of iron
- 4.53×10^{-3} kg of tin
- 2.99×10^{-3} kg of silicon

Adding Qwen3 as a foundation (Qwen3-0.6B-Base, Qwen3-1.7B-Base, Qwen3-4B-Base)

- 105.63 kg of copper
- 3.49 kg of iron
- 1.56 kg of tin
- 1.03 kg of silicon
- 0.85 kg of nickel

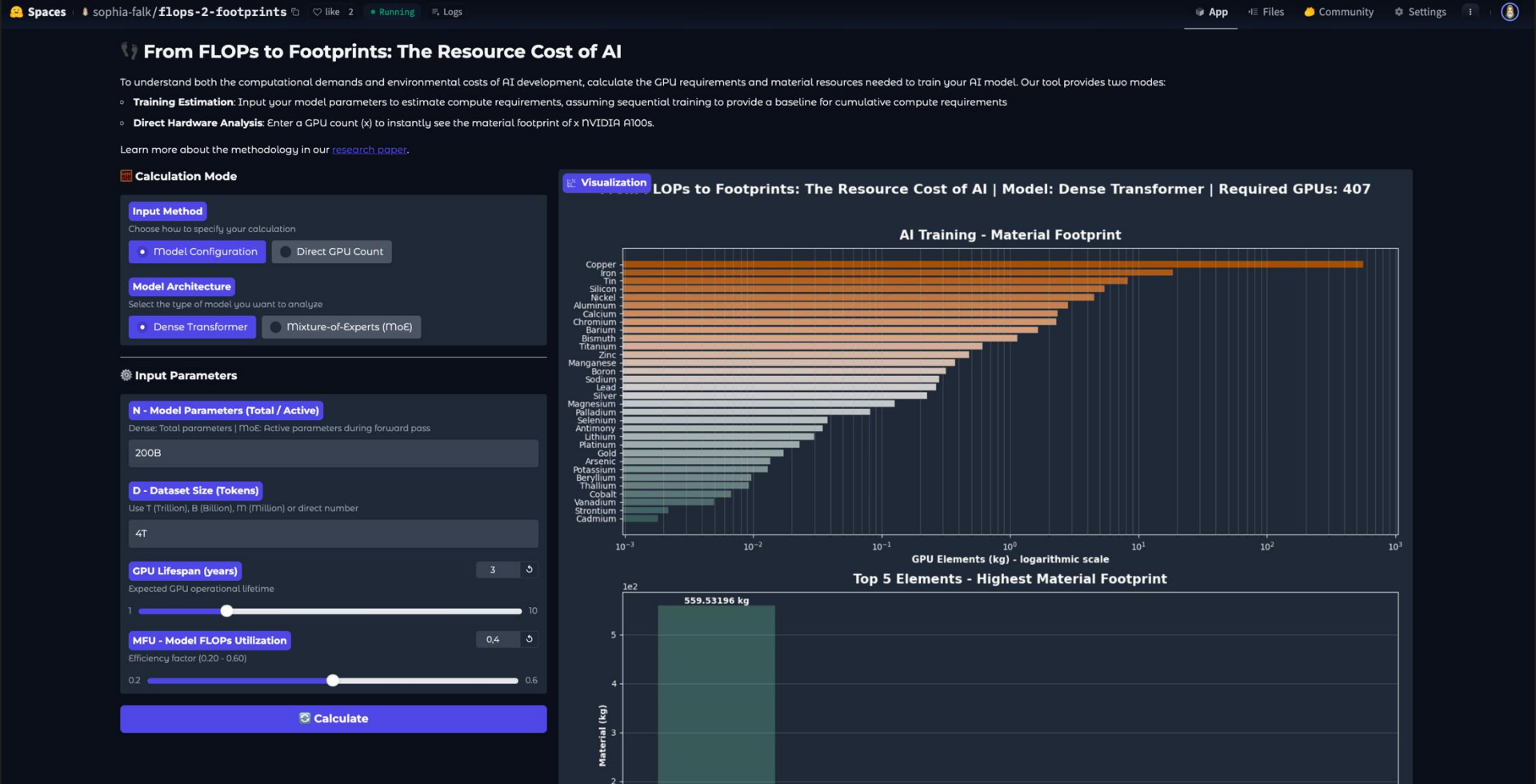
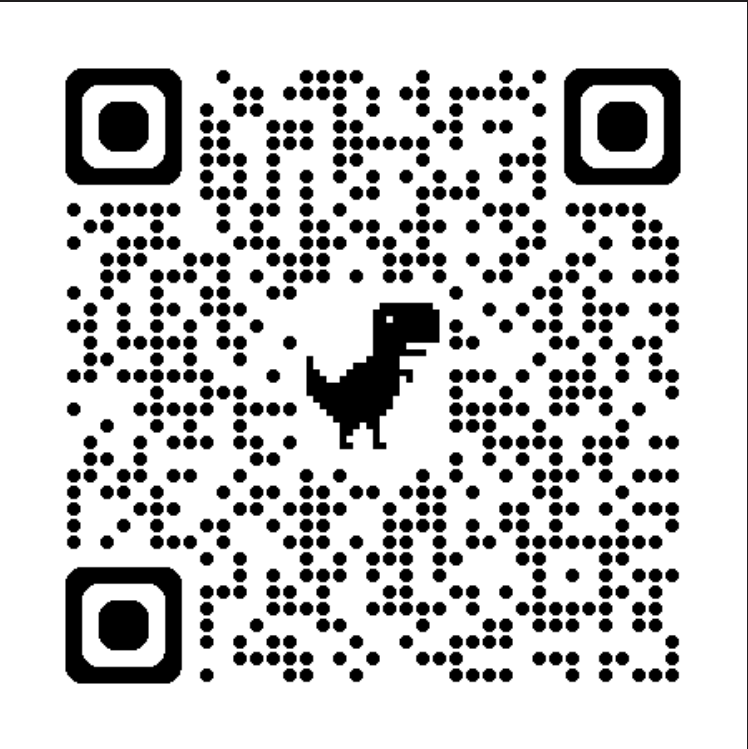
Table 9: Model configurations and training budgets (tokens) used for resource footprint estimation.

Family	Params	Base	Instruct	Think
Tucano2-0.6	670M	408B	—	—
Tucano2-qwen-0.5	490M	50B	4.5B	250M
Tucano2-qwen-1.5	1.5B	100B	4.5B	250M
Tucano2-qwen-3.7	3.7B	50B	4.5B	250M



<https://arxiv.org/html/2603.03543v1>

LINK TO WEBSITE





NEXT ON THE AGENDA



PICK UP FROM LAST STOP

Nicholas Kluge & Aniket Sen
Polyglot Team



LUNCH TIME



LLMS FOR HUMANITIES

14:00 — 15:00

Alexander Ermakov

Bonn Center for Digital Humanities (BCDH)



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